

TECHNICAL REPORT · CONFIDENTIAL

SEO & GEO Change Audit

A complete, file-level breakdown of the search and generative-engine optimization work on the `camillo` branch — what changed, what it was before, and how each change affects SEO and GEO.

Project

disruptorsmedia.com

Branch

camillo · 71 commits ahead

Prepared for

Head of SEO / GEO

Date

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Everything below lives on the **camillo** branch. **None of it is live until that branch is deployed** — so today the production site still behaves exactly as the "Prior" column describes. Read every "Prior → Now" pair as "live site → what ships on deploy."

0 The core problem this work fixes

PRIOR (LIVE TODAY)

Client-rendered React SPA. The server returns an empty shell (`<div id="root"></div>` + JS bundle); all content is painted by JavaScript after load. AI crawlers and render-budget-limited search crawlers saw a blank page. A ~1.8 MB entry bundle and heavy hero videos produced very poor LCP. Net: **~0% real content in the initial HTML.**

NOW (ON DEPLOY)

Indexable marketing routes are **prerendered to static HTML at build time** (SSR-equivalent snapshots), plus per-route metadata, structured data, and an AI-crawler access layer. Crawlers and LLMs get full text, titles, and schema in the raw HTML **without executing JS.**

1 Prerendering — the biggest single change

What: A build-time step (`scripts/prerender.js`) loads each marketing route in headless Chromium, lets it render and animate, then writes the fully-rendered HTML back to disk as `dist/<route>/index.html` . Netlify's catch-all redirect is non-forced, so these static files are served before the SPA fallback. The client still hydrates on top, so pages remain interactive.

Coverage: 23 core marketing routes + `/billboard` + **10 case-study pages** + **published blog posts pulled from Supabase at build time** (graceful fallback if DB creds are absent). Utility/app routes stay client-rendered on purpose.

SEO / GEO IMPACT

This is what makes content *exist* for non-JS crawlers and AI engines. It's the prerequisite for every other change in this report — titles and schema are worthless if the body is blank.

Two robustness fixes

- Each route renders in an **isolated browser context against a clean in-memory shell**, so one page's injected schema can't leak into another's snapshot.
- `emptyRoot()` forces the served shell's `#root` to be empty, so every route renders fresh rather than hydrating against stale markup — preventing a prior page's `<title>` /canonical from bleeding onto another route.

2 Per-route metadata

What: A lightweight head manager (`src/hooks/usePageMeta.js`) sets a unique `<title>` , meta description, `rel=canonical` , and Open Graph / Twitter tags per route — applied across all indexable routes, the 9 solution pages, all 10 case studies, and blog posts.

PRIOR
One static `<title>` and one description for the entire SPA — every URL shared identical metadata.

NOW
Self-describing, length-appropriate title per URL (homepage trimmed to ~46 chars), unique descriptions, correct canonicals, and OG/Twitter tags.

SEO IMPACT
Eliminates site-wide duplicate titles/descriptions (an explicit Screaming Frog finding), prevents duplicate-URL dilution via canonicals, and improves link-preview CTR on shared links.

3 Structured data (schema.org JSON-LD)

Schema type	Where & why
Organization, LocalBusiness, ProfessionalService, WebSite	Global entity definition (site-wide in <code>index.html</code>).
Service	On solution pages — defines each offering as a typed service.
FAQPage	FAQ page + all 9 solution pages — heavily used in AI Overviews.
HowTo + Person (author)	On solutions content — process + authorship signals.
Review	Real, attributable testimonials only — no fabricated <code>AggregateRating</code> (deliberately omitted to avoid structured-data spam).
BreadcrumbList	Case studies and deeper pages — SERP display + crawl-path clarity.
Article	Blog posts — <code>headline</code> , <code>datePublished</code> , <code>author</code> (Person), <code>articleSection</code> .

PRIOR
Effectively no structured data reached crawlers — it was JS-injected into an empty shell.

NOW
Full schema graph in the raw prerendered HTML.

SEO / GEO IMPACT
Schema is how AI engines and search resolve the brand as a known *entity* and understand page type and relationships. FAQ/HowTo are favored by answer engines. The "real reviews only" choice keeps E-E-A-T trust signals clean.

4 GEO / AI-crawler layer

- **llms.txt** added at site root (per the llms.txt spec): a structured, link-annotated map of core pages, all services, and resources, plus a one-paragraph entity summary. Validates clean.
- **robots.txt:** AI crawlers (GPTBot, ClaudeBot, PerplexityBot, OAI-SearchBot, Google-Extended) remain **allowed** (`User-agent: * / Allow: /`). The only change was **removing** `Crawl-delay: 1` , which was throttling crawl budget.
- **Answer-first content:** key passages restructured so the first sentence directly answers the implied question — the format AI engines preferentially extract and quote.

PRIOR

No `llms.txt` ; a crawl-delay slowing crawlers; marketing-prose-first content, not answer-first.

NOW

`llms.txt` map, unthrottled crawl, and answer-first passages.

GEO IMPACT

`llms.txt` + answer-first structure directly raise *citability* — the likelihood an LLM quotes or recommends the site. This is the lever that moves the stated 0%-AI-citation problem.

5 Case-study pages (most recent work)

What: All 10 case studies (`/work-segpro` , `/work-tradeworx-usa` , ...) were added to the prerender list and `sitemap.xml` , and each got `usePageMeta` with a unique title, a results-driven meta description, and BreadcrumbList schema. Their statistic-rich bodies (e.g. **+450% campaign ROI**, **-85% setup time**, **+300% lead quality**) now render into static HTML.

PRIOR

These routes existed but were **not prerendered and not in the sitemap** — crawlers/AI got the empty shell, so the hard numbers (the most citable content on the site) were invisible.

NOW

Prerendered, in the sitemap, with unique metadata and breadcrumbs; numeric results live in the raw HTML.

SEO / GEO IMPACT

Case studies are the highest-citability content an agency has — concrete, attributable, numeric results. Surfacing them in raw HTML is a direct citability and long-tail-keyword win.

Caveat to relay: in the *local* build these pages currently still show the homepage title, because the dev machine's build is disk-blocked and the compiled chunks are stale (they predate the `usePageMeta` edits). The **source is correct**; a clean build on the deploy pipeline produces the correct per-page titles + breadcrumb schema. This is a local-environment artifact, not a code defect.

6 Core Web Vitals / performance

Area	Change
Video	Homepage skips background videos on mobile and defers below-the-fold video; billboard hero video replaced with an optimized image; <code>FastVideo</code> lazy-loads via intersection observer.
Images	Marketing logos/images served via Supabase transform URLs (auto-resize + auto-WebP); 37 orphaned PNGs (~89 MB) deleted; broken Cloudinary URLs migrated to Supabase.
CLS	Explicit dimensions added to header logo, DualCTABlock background, footer logo, and decorative images.
Loading screen	Full-screen "0% — your experience is loading" gate and a spinner flash removed; <code>useIsMobile</code> made synchronous on first render to avoid a layout flash.
Service worker	Switched to network-first for HTML navigations so it can't pin stale content/markup.

SEO IMPACT

Direct LCP and CLS improvements (ranking + UX signals); lighter pages improve crawl efficiency; the service-worker fix stops returning visitors being stuck on stale cached pages after deploys.

7 Redirects, 404, sitemap, indexing control

- **sitemap.xml** rebuilt to 33 static entries (core + billboard + 10 case studies), with blog posts appended at build time.
- **Legacy 301 redirect scaffold** (`public/_redirects`) + a branded static `404.html` and `/404` route.
- `/free-business-audit` → `/marketing-audit` 301 added (fixed a broken internal link).
- `/event-checkin` marked `noindex, follow` via `X-Robots-Tag` in `netlify.toml` (utility page kept out of index and prerender).
- Over-length titles/descriptions trimmed; duplicate-title and broken-link Screaming Frog findings resolved.

PRIOR

Sitemap missing case studies; no legacy-redirect map (backlinks to old URLs hit soft-404s); generic SPA 404; thin indexing control.

NOW

Complete sitemap, redirect scaffold, branded 404, and explicit `noindex` on utility pages.

SEO IMPACT

Better discovery, backlink-equity recovery once the legacy map is filled, no index bloat from utility pages, and clean crawls.

→ Still needs a human (flagged, not done in code)

Item	Why it needs a person
Deploy <code>camillo</code> with the prerender wired into CI HIGHEST IMPACT	Nothing in this report is live until this happens. Infra — Bryan.
Populate the legacy 301 map	Scaffold exists; the real old→new URL pairs need human knowledge of old URLs.
Re-encode hero videos	Modern codecs/bitrates for further LCP gains.
Inject statistics / unique data into cornerstone pages	Audit measured citability passages as strong on structure but zero on statistical density / uniqueness — the remaining content lever.
Off-site / brand authority (Wikipedia, Reddit, LinkedIn entity presence)	Can't be done in this repo.

✓ Honest status note

Local verification is partially blocked: the dev machine has been near-full on disk, so a clean local build to screenshot final output wasn't completed, and there are **uncommitted changes** still in the working tree (the 10 case-study `usePageMeta` edits, blog `Article` schema, `prerender.js` blog-fetch + `emptyRoot`, `robots.txt`, `sitemap.xml`). The code is correct and reviewable; it needs a healthy environment to build, commit, and deploy. Everything already committed in sections 1–7 takes effect the moment the branch builds and ships.